

Win by Hook or Crook? Self-Injecting Favorable Online Reviews to Fight Adjacent Rivals

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Received: March 22, 2023

Revised: December 13, 2023; April 17, 2024;
August 1, 2024

Accepted: August 30, 2024

Published Online in Articles in Advance:
September 18, 2024

<https://doi.org/10.1287/isre.2023.0179>

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Abstract. The existing literature has long assumed that unethical behaviors are fueled by competition. Under this premise, rivalry and competition are often treated interchangeably. This study seeks to challenge this prevailing notion by introducing a conceptual distinction between rivalry and competition. Specifically, we explore the multifaceted relationship between rivalry and unethical behaviors in the context of online review self-injecting behaviors. Drawing on the collective regulatory focus framework and deterrence theory, we posit that rivalry-induced deterrence triggers a shift in mindset from promotion focus to prevention focus, which leads to the formation of mutual forbearance in self-injecting behaviors. Mutual forbearance collapses when one party unilaterally engages in self-injection, prompting the other party to retaliate by increasing their self-injecting activities. Additionally, we propose that firm reputation plays a crucial role in reinforcing the formation of mutual forbearance while mitigating its collapse. Employing an exhaustive data set from Ctrip.com and Qunar.com that comprises 85,939 observations from 3,996 hotels and spans 2018–2020, we found that the presence of an additional rival leads to a 3.2% decrease in self-injecting intensity. Furthermore, self-injecting intensity increases by 0.095% for every 1% rise in that of the rivals and increases by 0.041% for a 1% rise in that of the nonrival competitors. Crucially, our study identifies that firm reputation acts as a significant factor that strengthens the negative relationship between the number of rivals and self-injecting intensity and weakens the positive relationship between a hotel's self-injecting and its rival's self-injecting intensity. These empirical findings are supported by a myriad of robust examinations, including the use of a set of subsample analyses, alternative measures of independent variables, alternative instrumental variables, alternative observation time windows, and multiple endogeneity tests. This study contributes to the literature by highlighting the pervasive and potent role of rivalry as an understudied inhibitor of self-injecting behaviors. Our empirical evidence offers critical managerial implications for platforms, sellers, and consumers.

History: Juan Feng, Senior Editor; Xitong Li, Associate Editor.

Funding: This work was supported by the National Natural Science Foundation of China [Grants 72171189, 71901172, and 72131005] and the 111 Project [D21007].

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/isre.2023.0179>.

Keywords: rivalry • competition • unethical behavior • review manipulation • self-injecting favorable reviews

1. Introduction

Studies have long asserted that competition galvanizes firms to engage in unethical practices, such as cheating, lying, and sabotage (Chui et al. 2021, Vadera and Pathki 2021, Thanassoulis 2023). This assertion has been extensively corroborated in the context of online transactions, as multiple studies have found that hotels tend to self-inject more favorable online reviews in response to heightened competition (Lappas et al. 2016, Luca and Zervas 2016, Nie et al. 2022). The underlying

rationale posits that competition not only intensifies incentives for unethical actions but also fosters risk-taking behaviors (To et al. 2018). Notably, in these studies, competitors are often viewed as homogeneous economic actors vying for the same scarce resources.

However, not all competitors are alike. In certain instances, competitors, as exemplified by the iconic dynamic between Magic Johnson and Larry Bird¹ in basketball, embody profound relational elements characterized by mutual recognition and respect (Kilduff

2019, Ager et al. 2022). Incorporating this relational view into the realm of competition, a recent body of research has coined the term “rivalry” to denote this unique form of competition (Kilduff 2019, Sgourev and Operti 2019, Sharapov and Ross 2023). Rivalry, as defined by Kilduff et al. (2010), refers to the competitive relationship between a focal firm and a target firm, distinguished by the focal firm’s experience of heightened psychological stakes of competition when competing against the target firm. Importantly, studies advocating a positive link between competition and unethical behaviors have starkly conflated rivalry with competition, leaving us a significant paucity in our understanding of how rivalry specifically influences firms’ self-injecting behaviors. In essence, rivalry is a concept related to but different from competition, as it possesses a myriad of unique idiosyncrasies (Yip et al. 2018, Operti et al. 2021). First, rivalry involves relational factors. Unlike competition, which can take place between unfamiliar or anonymous opponents, rivalry concentrates on a specific, identifiable, and familiar opponent (Kilduff et al. 2010, 2016). Second, the essence of rivalry extends beyond the current state of conflicting goals; it also encompasses historical interactions (Jarzabkowski and Bednarek 2018). Akin to the development of friendship, rivalry typically evolves over time through recurrent interactions. These historical interactions between rivals facilitate the establishment of mutual recognition and respect. Third, while competition may involve unevenly matched contests, where one competitor may potentially be superior to the other(s), rivalry inherently entails evenly matched contests (Pike et al. 2018).

We contend that the presumed positive link between competition and unethical behaviors may not hold in the case of rivalry due to its unique characteristics. On the one hand, as rivalry represents an evenly matched contest and unethical behaviors are more likely to occur in such scenarios (To et al. 2018, Vadera and Pathki 2021), it is plausible to expect that, akin to competition, rivalry would positively predict self-injecting behaviors. On the other hand, because rivalry centers on a specific, identifiable, and familiar opponent, firms are interested in and capable of scrutinizing the unethical behaviors of their rivals, thereby leading to self-restraint in engaging in unethical actions (Sun 2021). Moreover, the presence of mutual recognition and respect in rivalry diminishes one’s inclination to outperform their rivals through unfair means. These inspiring studies imply that, in contrast to competition, rivalry may serve as a deterrent to unethical behaviors. Furthermore, given the considerable heterogeneity among firms in their behaviors involving unethical actions, it is improbable that the influence of rivalry on firms’ online self-injecting behaviors would be uniformly applied across hotels with heterogeneous attributes. These conflicting views therefore motivate us to advance this line of

research and explore the following questions in greater depth.

Research Question 1. *Does rivalry, like competition, fuel online self-injecting behaviors or inhibit online self-injecting behaviors?*

Research Question 2. *How does the effect of rivalry on firms’ online self-injecting behaviors vary with hotel heterogeneity?*

To address the above questions, we first conceptually differentiate rivalry from competition. Based on the theoretical definition of rivalry, we then develop a novel measurement method, including both objective filtration and subjective evaluations, to further discern rivals from competitors. Embracing collective regulatory focus framework and deterrence theory, we seek to explore the relationship between rivalry and self-injecting behavior from a dynamic point of view. In organizational settings, firms typically assemble teams to deal with review-related decisions. Through ongoing interaction, review management team members cultivate a shared understanding of the values of the team to regulate or guide their behaviors toward achieving collective goals (Owens and Hekman 2016). The collective regulatory focus theory suggests that teams are motivated by either a promotion focus, aimed at maximizing gains, or a prevention focus, focused on minimizing losses to guide their collective behaviors (Dimotakis et al. 2012, Emich and Vincent 2020). Because self-injecting favorable online reviews entails both gains (i.e., high rating, increased web traffic) and losses (i.e., reputation damage, potential penalties), To et al. (2018) illuminate that rivalry promotes a strong desire to outperform and leads to a promotion mindset. Recent studies have provided supporting evidence that a promotion mindset increases risk-taking behaviors, which fuels self-injecting intensity (Ager et al. 2022). Deterrence theory, on the other hand, posits that sanctions or punishments could serve as a deterrence mechanism against unethical behaviors (Yu et al. 2009). Because of the historical and relational nature of rivalry, firms possess the capability to monitor and track the actions and strategic maneuvers of their rivals (Uribe 2020). If a hotel self-injects favorable online reviews, these unethical behaviors are likely to be observed by rivals, who may then report to regulators, thereby increasing the likelihood of punitive measures. Thus, we argue that rivalry-induced deterrence contributes to the formation of mutual forbearance in acting unethically, which hinders self-injecting intensity. As the number of rivals increases, the deterrence against acting unethically also intensifies. Consequently, two competing conjectures emerge: the number of nearby rivals would either *foster* or *impede* self-injecting behaviors.

Further, mutual forbearance signifies an implicit agreement between a focal firm and its rivals, wherein all parties refrain from engaging in the practice of self-injecting favorable reviews. Additionally, the unilateral betrayal of this agreement manifested through observing a rival's self-injecting behaviors serves as a signal of the collapse of mutual forbearance in the rivalry. The breakdown of mutual forbearance triggers a retaliatory reaction against the betrayers in a more intensive manner (Ryu et al. 2020). Intuitively, another vividly portrayed conjecture emerges: the self-injecting behaviors of rivals prompt the focal hotel to engage in self-injecting to a greater extent than nonrival competitors. The primary cost of self-injecting behavior lies in the potential loss of reputation when such behaviors are exposed. Insights from organizational studies suggest that reputable firms tend to suffer more from scandals than less reputable counterparts due to negative spillovers resulting from firm misconduct (Knittel and Stango 2014, Johnson et al. 2018, Paruchuri et al. 2021). Given this, it can be argued that rivalry-induced deterrence would be more pronounced for reputable hotels than for less reputable ones. Consequently, we conjecture that the relationship between rivalry and self-injecting intensity would be influenced by the reputation of the hotel.

We empirically examined the aforementioned hypotheses using comprehensive panel data sets from [Ctrip.com](#) and [Qunar.com](#), which encompassed 85,939 observations from 3,996 hotels and spanned the period from April 1, 2018, to November 30, 2020. The examination of the impact of rivalry on firms' online self-injecting behaviors poses challenges vis-à-vis potential endogeneity issues. To mitigate this concern, we employed an instrumental variable approach. Furthermore, we conducted a series of supplementary tests to rule out alternative explanations and enhance the robustness of our findings. Our analyses reveal compelling evidence that the presence of nearby rivals acts as a deterrent to a hotel's self-injecting intensity. More precisely, each additional nearby rival corresponds to a 3.2% decrease in a hotel's self-injecting intensity. Comparative analyses underscore that a hotel's online self-injecting intensity is more responsive to the self-injecting behavior of nearby rivals than to that of nearby nonrival competitors. Specifically, self-injecting intensity increases by 0.095% for every 1% rise in the intensity of rivals and by 0.041% for a 1% rise in the intensity of nonrival competitors. In addition, our investigation reveals that hotel reputation plays a pivotal role in strengthening the negative relationship between the number of rivals and a hotel's self-injecting intensity while weakening the positive relationship between the hotel's self-injecting intensity and its rival's self-injecting intensity. These interesting results challenge the long-standing assertion that competition fuels unethical behaviors and shed new light on the multifaceted nature of rivalry in the evolving context of online

reviews (Vadera and Pathki 2021, Chang and Sokol 2022, Nie et al. 2022).

This study makes several significant contributions to theory. First, this study ushers in a novel conceptual distinction between rivalry from competition in the accommodation sector in general and the context of hotel online reviews in particular and further advances our understanding of the epistemological roots of online self-injecting behaviors by pinpointing rivalry as a previously unexplored inhibitor of such unethical behavior. Second, this study quantifies the effects of rivalry, both in terms of the number of rivals and the intensity of rivalry, on firms' online self-injecting intensity. Third, this study uncovers the multifaceted effect of rivalry on unethical behaviors from a dynamic point of view. Fourth, this study sheds new light on how the influence of rivalry on firms' online self-injecting behaviors differs from that of nonrival competition, thereby providing novel insights into the conditions under which rivalry is generally more or less likely to lead to firms' self-injecting behaviors. Lastly, this study enriches the literature on rivalry by demonstrating how firm reputation can foster the positive aspects and circumvent the negative consequences of rivalry.

2. Related Literature

2.1. Self-Injecting Favorable Online Reviews in the Accommodation Sector

The global accommodation market has witnessed increasing dominance by a handful of well-established reservation platforms, such as [Booking.com](#), [TripAdvisor](#), [Expedia](#), and so on (Lappas et al. 2016). When travelers plan a trip, these online reservation platforms play a paramount role as users usually search for hotels on them. Admittedly, the visibility of a hotel in search results on these platforms significantly influences consumer choices. Studies have shown that websites listed on the first page of search engine results receive almost 95% of web traffic, leaving only 5% for those on subsequent pages of search results (Kaye 2013). Although consumers increasingly rely on online reviews to assess the quality of products prior to making purchasing decisions (Feng et al. 2019, Kwark et al. 2021, Fang 2022), the online reservation market remains highly concentrated. As hotels vie for attention on these renowned platforms, securing top-ranking positions becomes crucially important for hotel managers seeking to attract more web traffic. To influence consumer perceptions and promote their rankings, firms often resort to self-injecting favorable consumer reviews of their products on their websites, review platforms, and e-marketplaces² (Mayzlin et al. 2014, Luca and Zervas 2016).

Self-injecting favorable online reviews have been proven to corrode review credibility (Jabr 2022), cause rating bias (Hu et al. 2017), and diminish consumer

surplus (Mayzlin et al. 2014). From a pragmatic perspective, e-platforms such as Amazon and Yelp have responded to this challenge by implementing opinion spam algorithms to detect and filter fake reviews. Further, from an epistemological perspective, a wealthy line of studies has examined the relationship between product characteristics and self-injecting behaviors. For example, Hu et al. (2012) suggest that experience products tend to attract more fake reviews than search products on Amazon. Sellers frequently self-post positive online reviews of newly launched products because early consumer reviews of a product are crucial to its success (Park et al. 2021). A large number of empirical studies have shown that existing positive evaluations tend to attract further positive evaluations while existing negative evaluations attract less consumer attention (Goes et al. 2014, Li 2018, Schneider et al. 2021). Park et al. (2021) suggested that a product has a higher average rating and a greater review volume after a positive initial consumer review compared with a negative one. In addition, firm heterogeneity has also been found to be correlated with firms' self-injecting behaviors. For example, Mayzlin et al. (2014) found that chain hotels are less likely to engage in self-injecting behaviors than nonchain hotels. Further, Luca and Zervas (2016) suggested that firms are more prone to commit online review fraud when their reputation is weak. This might occur when firms have few reviews, have recently received bad reviews, or possess a low average rating.

The economics and psychology literature has long recognized competition as a catalyst for unethical behaviors (Lee et al. 2018, Chui et al. 2021, Vadera and Pathki 2021). Because self-injecting favorable online reviews constitutes a typical form of unethical business practices, a fruitful line of research has positioned competition as one of the primary causes of firms' self-injecting behaviors. For example, Mayzlin et al. (2014) found that the geographic density of competitors increases the likelihood of firms' engagement in self-injecting behaviors. Moreover, Lee et al. (2018) showed that the growth of self-injecting behavior might be attributed to escalating competition. Although certain competitors or rivals can instill a motivation to perform beyond an ordinary competitive spirit or the objective stakes of the contest, prior studies have often treated rivalry and competition as interchangeable terms, resulting in a dearth of knowledge about the specific effect of rivalry on firms' self-injecting behaviors. Furthermore, prior studies have yet to identify the conditions under which the relationship between competition and self-injecting behavior is more or less likely to manifest. Thus, there is a paucity of understanding vis-à-vis whether any rivalry exists among the competitors being studied, which might confound the effects of rivalry and nonrival competition on self-injecting behaviors. This study diverges from prior research by conceptually and empirically differentiating

rivalry from nonrival competition and comparing their effects on firms' online self-injecting behaviors.

2.2. Differentiating Rivalry from Competition

Competition is ubiquitous in life, from employees vying for promotions to athletes striving for championships. Existing literature predominantly holds a structural view of competition and views it as a situation in which the outcomes of actors are diametrically opposed (Bennett et al. 2013, Charness et al. 2014). However, competitors are not homogeneous per se. As Kilduff et al. (2010) suggested, "certain competitors, or rivals, can instill a motivation to perform that goes above and beyond an ordinary competitive spirit (p. 943)." For example, two soccer teams in the England Premier League, Manchester City and Manchester United, consider their contests against each other (known as "Local Derby") as considerably more important than games against other teams, reflecting an obsessive focus on their relative performance.

In prior competition literature, the term "rivalry" has been generally used synonymously with the term "competition"; rivals are simply opponents in competition with one another in a competitive setting (Garcia et al. 2006, Luca and Zervas 2016, Kitchens et al. 2018, Chan and Briers 2019, Chui et al. 2021, Vadera and Pathki 2021). However, recent research (Kilduff et al. 2016, To et al. 2018, Operti et al. 2021, Ager et al. 2022) challenges this perspective and suggests that rivalry encompasses more than a mere current state of conflicting or opposing goals. Rivalry exists when the psychological stakes of competition increase as a result of the existing relationship between competitors, and it is independent of objective stakes or other structural or situational characteristics (Kilduff et al. 2016). In essence, there are two types of competitive influence: one driven by rivals and the other by nonrival competitors (Kilduff et al. 2016, Connelly et al. 2019). The key difference between rivalry and nonrival competition lies in the historical and relational nature of rivalry.

Importantly, researchers suggest that "rivalry is a relational schema in which the representations of partner, self, and dyad are linked to representations (i.e., memories, and expectations) of shared competitive interactions" (Converse and Reinhard 2016, p. 191). Therefore, rivalry is neither a single competition nor its immediate outcomes; however, it is psychologically embedded within an ongoing competitive narrative with relational components. The conceptualization of rivalry likens to friendship: a relationship that typically evolves from repeated social interaction (Kilduff et al. 2016). In contrast to nonrival competition that can take place between unknown or anonymous opponents, rivalry entails a focus on a specific, identifiable opponent (Kilduff et al. 2010). Conversely, given that rivalry represents a relationship, it is always directed toward a

recognized competitor, featuring a historical component built up over time through recurrent interaction. The full scope of rivalry cannot be captured by the current state of conflicting goals. To better illustrate the differences between rivalry and nonrival competition, we present the following example of Hilton and Marriott in the accommodation sector. In their pursuit of being world-leading chain brands, Hilton is one of Marriott's well-known rivals, engaging in historical, evenly matched competitions. Hotels of these two chain brands have similar price ranges, offer comparable facilities, and are situated in close proximity, competing for each other's consumers. On the other hand, there are other well-known brand chain hotels such as Holiday Inn, which are considered nonrival competitors to Marriott.

3. Theoretical Underpinnings and Hypothesis Development

In the context of online review ratings, we meticulously embrace the collective regulatory focus framework and deterrence theory to formulate specific predictions on how rivalry influences online self-injecting behaviors. In particular, we focus on rivalry-induced deterrence to elucidate why rivalry differs from competition in predicting online self-injecting behaviors. Additionally, we delve into the nuanced aspect of how the influence of rivalry is contingent on the heterogeneity of hotels. The theoretical rationale underlying our hypotheses is foreshadowed in Figure 1.

3.1. Collective Regulatory Focus Theory and Risk-Taking in Rivalry

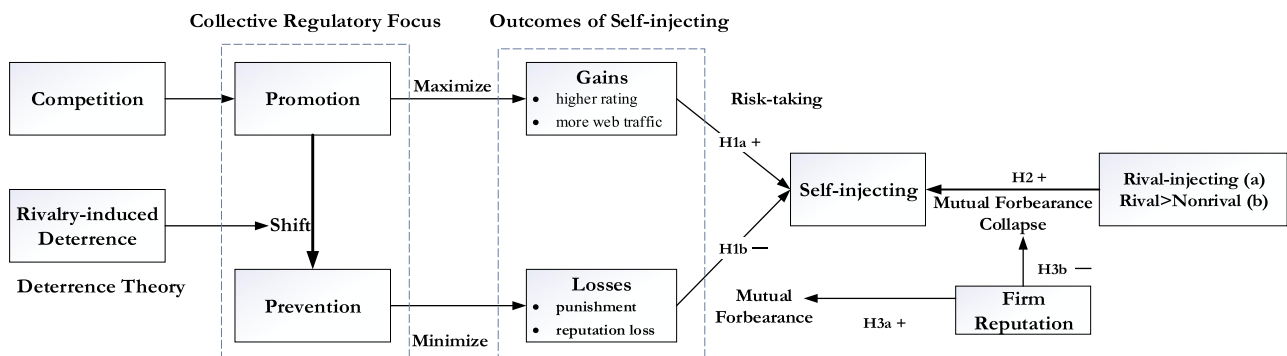
In this section, we seek to explore the relationship between rivalry and a hotel's self-injecting behavior through the lens of collective regulatory focus theory. The practice of self-injecting favorable reviews entails both positive and negative outcomes for hotels. Because hotels typically assemble teams to deal with review-related decisions, the collective regulatory focus theory provides a pertinent lens through which we can comprehend this phenomenon, suggesting that teams

employ two distinct means of self-regulation, known as promotion focus and prevention focus, to guide their collective behaviors (Faddegon et al. 2008, Owens and Hekman 2016). Promotion focus emphasizes the pursuit of gains (or the avoidance of nongains), whereas prevention focus focuses on the avoidance of losses (or the pursuit of nonlosses). According to collective regulatory focus theory, teams are driven by either a promotion focus or a prevention focus. Those exhibiting a promotion focus tend to be more attuned to positive outcomes, harbor stronger nurturance needs, care more about their accomplishments, and pay closer attention to the results of gains or nongains. Conversely, teams with a prevention focus are more attuned to negative outcomes, possess stronger security needs, care more about their responsibilities, pay more attention to the results of loss or nonloss, and utilize avoidance as a strategic means of mitigation (Faddegon et al. 2008, Dimotakis et al. 2012).

Rivalry heightens the psychological stakes of competition and promotes a strong desire to outperform each other (Jarzabkowski and Bednarek 2018). When engaged in rivalry, people are more result-orientated than when competing against nonrival competitors (Kilduff et al. 2016). Thus, rivals propel each other toward higher levels of motivation and performance. Hotel's review management teams are more concerned about winning rather than the means to achieve victory when they compete against rivals, as rivalry elevates the subjective importance placed on competitive outcomes (Pike et al. 2018, Connelly et al. 2019, Operti et al. 2021). As rivalry increases the psychological stakes of competition, this heightened effort in the pursuit of maximum performance is consistent with a promotion mindset whereby teams seek to maximize gains.

Prior work has suggested that teams with a high promotion focus are more inclined to take risks because a focus on the pursuit of maximizing gains increases tolerance for risk (Bennett et al. 2013). Furthermore, Ma (2023) found that promotion-focused teams employ riskier strategies in pursuit of gains (e.g., pursuing more aggressive performance at the expense of reduced

Figure 1. (Color online) Theoretical Foundations of the Hypotheses



accuracy), whereas prevention-focused participants adopt more cautious strategies that prioritize avoiding losses (e.g., trading slower performance for increased accuracy). Collectively, these studies suggest that a rivalry-induced promotion mindset will lead to greater risk-taking behaviors because people commit themselves to maximizing positive outcomes. Although self-injecting is generally considered unethical and may incur punishment, we expect that hotels will take a risk to do so to outperform their rivals. As the number of rivals increases, the psychological stakes of competition heightens and the perceived ease of winning decreases (Chui et al. 2021). Previous studies suggest that people are more prone to engage in risk-taking behavior when the competitive outcome is attractive, and the likelihood of winning is low (Vadera and Pathki 2021). Hence, being in a larger (versus smaller) rivalry pool can fuel risk-taking tendencies and thus boost a hotel's inclination toward self-injecting behaviors. We therefore posit the following hypothesis.

Hypothesis 1a. *The number of nearby rivals positively influences a focal hotel's online self-injecting intensity.*

3.2. Rivalry-Induced Deterrence and the Formation of Mutual Forbearance

Integrating rivalry-induced deterrence theory into collective regulatory focus theory, this section provides an alternative view regarding the relationship between rivalry and a hotel's self-injecting intensity. Deterrence theory submits that sanctions or punishments could serve as a deterrence mechanism to deter deviant behavior (Yu et al. 2009). The efforts of sanction or punishment are measured by two dimensions: severity and certainty of punishment (Burns et al. 2023). Severity of punishment refers to the degree of punishment associated with not complying with security policy, and certainty of punishment pertains to the probability of being punished. Sanctions could manifest in either formal or informal (Pate and Hamilton 1992). Formal sanctions involve concrete penalties imposed by society (e.g., fines or imprisonment for criminal acts) or employers (e.g., demotions or employment termination for policy violations), whereas informal sanctions entail the disapproval from oneself, the community, or moral beliefs for a given action (Johnston et al. 2015).

Because of the historical and relational nature of rivalry, firms can track the actions and strategic maneuvers of their rivals, leading to a transparent competition among them (Uribe 2020). In contrast to novel or anonymous competitors, rivals represent counterparts with whom actors have long-standing relationships and engage in intensive interactions. The behaviors of rivals are more prone to observation and notice compared with those of nonrivals. In the event that a hotel acts unethically, such as self-injecting favorable online reviews, these

unethical behaviors could soon be detected by rival hotels. Consequently, firms may report the self-injecting behaviors of their rivals to authorities or platforms, escalating the likelihood of punishment for the rivals. Beyond formal sanctions, self-injecting favorable reviews may incur informal sanctions, such as disapproval of self and community. As discussed above, rivalry involves mutual recognition and respect (Ager et al. 2022), which may mitigate the inclination to win through unfair means. Competing against a rival can invoke greater concerns about one's own sense of self-worth, thereby rendering competition against rivals more threatening due to self-esteem considerations (Kilduff et al. 2016).

The deterrence imposed by a rival's formal or informal sanctions may shift a hotel's collective regulatory focus from a promotion focus to a prevention focus. For example, Gino and Margolis (2011) suggested that prevention focus can reduce unethical behavior by inducing a fear of punishment. Prior regulatory focus studies suggest that a prevention mindset curtails actors' appetite for risk by causing teams to pursue more familiar and conservative strategies as well as eschew novel or riskier strategies that may leave them vulnerable (Chakravarthy et al. 2021). We therefore expect that rivalry-induced deterrence contributes to the formation of mutual forbearance. In essence, mutual forbearance is a form of tacit collusion that alludes to the phenomenon of diminished aggressive actions between firms that compete against each other (Golden and Ma 2003). In such conditions, hotels might not act aggressively because the expected gains from self-injecting favorable reviews may be lower than the potential losses due to retaliatory measures from rivals. As the number of rivals increases, the deterrence from rival's formal and informal sanctions intensifies. This heightened deterrence leads to rivals viewing the payoff from mutual forbearance to be more attractive than the payoff from self-injecting favorable reviews. We therefore posit the following.

Hypothesis 1b. *The number of nearby rivals negatively influences a focal hotel's online self-injecting intensity.*

3.3. Collapse of Mutual Forbearance in Self-Injecting

On reservation platforms, the ranking for each hotel is inversely correlated with that of its rivals. The primary objective behind self-injecting favorable reviews is to increase hotel ranking and thus attract more web traffic. Hotel managers increasingly find themselves in a "prisoner's dilemma" scenario regarding whether to self-inject favorable reviews or not (Gossling et al. 2018). If a focal hotel and its rivals opt to self-inject at the same time and to the same degree, their relative ranking on the reservation platforms would remain unchanged, resulting in a lose-lose situation. Instead, the potential

win-win situation for this prisoner's dilemma scenario is that the hotel and its rivals established a tacit collusion of mutual forbearance, wherein none of them engages in self-injecting favorable reviews. Despite this, some hotels start to self-inject favorable reviews intensively as they anticipate obtaining a huge competitive advantage by unilaterally betraying the mutual forbearance agreement. Notably, in the accommodation sector, the observation of a rival's self-injecting behaviors serves as a signal of the collapse of mutual forbearance.

Previous studies indicate that the collapse of mutual forbearance can entail one retaliating against the betrayers (Chung and Beamish 2010, Ryu et al. 2020). In the context of online reviews, hotels possess various ways to retaliate against the betrayers, but the quickest, most effective, and most commonly used way of retaliation is to engage in more intensive self-injecting of favorable reviews³ (Nie et al. 2022). We therefore expect that the self-injecting behavior of the rival will trigger the focal hotel to retaliate in a similar manner. When observers perceive that rivals have gained desired benefits (e.g., higher ranking and more web traffic) or opportunities through self-injecting, negative emotions such as anger, envy, and a sense of injustice are often evoked. Research has consistently shown that these negative emotions are found to correlate with more aggressive retaliations in rivalry than in nonrival competition (Jones and Barnett 2022, Malik and Pichler 2023). Taken together, we propose that the self-injecting behaviors of rivals will lead the focal hotel to engage in more intensive self-injecting behaviors than nonrival competitors. Therefore, we put forth the following hypotheses.

Hypothesis 2a. *The self-injecting intensity of nearby rival hotels positively influences a focal hotel's online self-injecting intensity.*

Hypothesis 2b. *A hotel's online self-injecting intensity is more responsive to the self-injecting behavior of nearby rivals than to that of nearby nonrival competitors.*

3.4. Differential Effects of Firm Reputation on the Formation and Collapse of Mutual Forbearance

The primary cost of self-injecting behavior is the risk that the activity will be publicly exposed by its rivals and punished by the platforms or authorities. Admittedly, self-injecting behaviors could improve hotel rankings through numerical ratings and influence consumer decisions through textual persuasion. Government fines, penalties imposed by the review-hosting platform, and reputation damage are among the punishments that deter focal hotels from self-injecting favorable reviews (Mayzlin et al. 2014). However, the cost associated with self-injecting behaviors (i.e., reputational loss) is higher for reputable hotels than for less reputable ones. Organizational studies suggest that reputable firms ensure

more severe consequences from scandals than less reputable counterparts due to negative spillovers resulting from firm misconduct (Johnson et al. 2018, Paruchuri et al. 2021). For example, if a brand chain hotel is detected in self-posting favorable reviews, adverse publicity and potential sanctions could extend to all properties under the brand (Mayzlin et al. 2014). This phenomenon is also exemplified in a recent case, where the Irish hotel Clare Inn Hotel and Suites, part of the Lynch Hotel Group, was issued a "red badge" by TripAdvisor, warning consumers that the hotel manipulated reviews after it was discovered that a hotel executive solicited positive reviews. TripAdvisor also removed reviews from other Lynch Hotel Group hotels. In such cases, platforms take action against the entire hotel group, given the common ownership and management of the hotels. Therefore, we posit that rivalry-induced deterrence will be more prominent for reputable hotels than for less reputable counterparts. Mutual forbearance denotes the situation in which hotels reach a tacit collusion with their rivals. The establishment and sustainment of mutual forbearance in self-injecting behavior requires a foundation of mutual trust, as firms would get huge advantages if one unilaterally betrays the mutual forbearance (Nie et al. 2022). In this context, we propose that firm reputation plays a vital role in intensifying the formation of mutual forbearance and strengthening the negative relationship between the number of rivals and a firm's self-injecting intensity. Fundamentally, reputation is an observer's impression of the actor's disposition to behave in a certain manner (Clark and Montgomery 1998). Reputation signifies a firm's credibility and moral capital, given that reputation involves significant investments and accumulates over time, thereby suggesting a firm is unlikely to act otherwise (Dal Bo and Tervio 2013, Servaes and Tamayo 2013, Chiles 2021). Previous studies affirm that consumers tend to deposit more trust in reputable firms than in those with low reputation (Kim 2019). Hotels, in turn, may believe that reputable rivals are more likely to behave as agreed (not to self-inject) than less reputable rivals and mutual forbearance is more likely to be formed among reputable hotels. We expect that firm reputation reinforces the formation of mutual forbearance and posit the following.

Hypothesis 3a. *Hotel reputation strengthens the negative relationship between the number of nearby rivals and online self-injecting intensity.*

Further, observing a rival's self-injecting behavior may trigger the focal hotel to retaliate by engaging in more intensive self-injecting practices. We expect that such a positive link would be weakened by the influence of firm reputation. Compared with less reputable hotels, reputable hotels usually have a larger volume of reviews on reservation platforms (Hollenbeck 2017, Chiles 2021). Additionally, on most reservation platforms, the ranking of a hotel depends on its average

rating that is computed by the arithmetic mean value of all consumer reviews within specified periods (Ghose et al. 2012, Chambers and Baker 2020, Hunold et al. 2020). Therefore, the ranking of hotels with a larger quantity of consumer reviews is less susceptible to the impact of a certain number of fake reviews than that of hotels with fewer consumer reviews. In support of this argument, Chen and Papanastasiou (2021) suggested that firms tend to frequently self-inject positive reviews for products with few consumer reviews. Consumers often peruse previous reviews before making a purchase, and the persuasiveness of a self-posted favorable review on consumer decisions would be less impactful when many consumer reviews are available. Therefore, the effectiveness of self-injecting as a retaliatory strategy against a rival's self-injecting behavior is lower for reputable hotels than for less reputable counterparts. Moreover, the reputation loss incurred from unethical behaviors is higher for reputable hotels than for less reputable ones (Mayzlin et al. 2014, Luca and Zervas 2016). Because the effectiveness of self-injecting as a retaliation to a rival's betrayal of mutual forbearance is lower and the risk is higher for reputable hotels than for less reputable ones, we expect that hotel reputation weakens the positive relationship between self-injecting behaviors of rivals and the focal hotel's self-injecting intensity.

Hypothesis 3b. *Hotel reputation weakens the positive relationship between the self-injection of rivals and online self-injecting intensity.*

4. Data

4.1. Sample

To empirically validate the aforementioned hypotheses, we collected hotel reviews from two leading reservation platforms in China, namely [Ctrip.com](#) and [Qunar.com](#). We chose to focus on the hotel category on [Ctrip.com](#) and [Qunar.com](#) to construct our data set for three reasons. First, competition, especially rival competition, is ubiquitous in the accommodation sector, making it an ideal research context (Chiles 2021). Second, because a hotel is a typical experience product, consumer reviews play a crucial role in shaping travelers' decisions (Ghose et al. 2012, Lappas et al. 2016). Moreover, prior research has indicated that review manipulation is more prevalent in experience products than in search products, making the hotel category a compelling context for examining the relationship between rivalry and review manipulation (Hu et al. 2011). Third, [Ctrip.com](#) and [Qunar.com](#) stand out as leading reservation platforms in China, and they are well recognized for their extensive collection of hotel reviews (Chen et al. 2019). These online platforms provide a natural field for experimentation to identify review manipulations due to their distinct review

policies. [Ctrip.com](#) is a "closed" reservation platform, meaning that only consumers who have stayed at a hotel can post a review for that specific hotel. On the other hand, [Qunar.com](#) is an "open" accommodation forum in which anyone can publish a review of any hotel. This means that publishing self-injecting positive reviews on [Qunar.com](#) is considerably less costly for a hotel than publishing positive ratings on [Ctrip.com](#). When hotels seek to self-post favorable consumer reviews, a rational choice is to do so on platforms with an "open" review policy, such as [Qunar.com](#). Empirical work by Mayzlin et al. (2014) has confirmed this conjecture. In line with the existing literature, we quantified firms' self-injecting intensity by comparing the differences in the distributions of positive reviews for the same hotel on [Qunar.com](#) and [Ctrip.com](#) (Mayzlin et al. 2014, Zhuang et al. 2018, Nie et al. 2022).

Given the extensive volume of online hotel reviews spanning different travel destinations, we chose the city of Shanghai as our focal point: a major metropolitan area and a popular destination for both business and leisure travelers in China. Unlike smaller cities or resort locales, a major metropolitan city such as Shanghai has a wide variety of hotels catering to various traveler preferences. In addition, a cosmopolitan hub such as Shanghai encompasses different competing modes, including both rival and nonrival competitions, which are the central focus of the current study. We developed a Python-based crawler for data collection. The crawler first retrieved the data of all available hotels in Shanghai. The names of the hotels as well as their contact phone numbers, locations, and postal code were collected. This process resulted in 12,975 hotels listed on [Ctrip.com](#) and 10,582 hotels listed on [Qunar.com](#) in Shanghai, respectively. We intended to compare the review distributions of the same hotel listed on both [Ctrip.com](#) and [Qunar.com](#). Because some hotels were not listed on either site, we developed a matching algorithm to identify the hotels that were listed on both sites to enable a robust comparison. The algorithm matched hotels based on *postal code*, *name*, and *address*. We then manually verified the matched list to ensure accuracy in the matching process. Through this combined matching approach, we identified 8,633 hotels located in 135 postal code areas available on both sites in our sample. Because we are interested in the distribution of consumer reviews, 427 hotels that had no consumer reviews on either [Ctrip.com](#) or [Qunar.com](#) throughout our observation period were excluded from our sample. We also dropped 1,051 hotels that commenced operations after the beginning of our database in April 2018. We further abandoned 174 hotels as they had been in business for less than six months. After these initial sample refinement processes, our sample comprised 6,981 hotels for further analysis. To enhance data accuracy, we then conducted telephone interviews

with the managers of these 6,981 hotels one by one. After three rounds of telephone interviews, the final sample contained 3,996 hotels.⁴ Heckman’s two-stage models were employed to ascertain that there is no self-selection bias in our sample, following the methodology proposed by Clougherty et al. (2016).⁵

For each of these 3,996 hotels in our study, we collected all the historical reviews, including the identity of the reviewer (as displayed by the site), review date, review ratings (from one to five), review content, the presence of management response to the review, and the publication date of the management response on both [Qunar.com](#) and [Ctrip.com](#). To capture the temporal dynamics of hotel ratings in different periods, we used one month as the time window and calculated the monthwise average hotel rating. We also collected hotel-specific characteristics, such as hotel star ratings, hotel locations, hotel ages, and a series of hotel service dummies (e.g., fitness, restaurant, and swimming pool). Overall, we collected monthly data on the hotels from [Qunar.com](#) and [Ctrip.com](#) for 32 months, from April 1, 2018, to November 30, 2020. Our final unbalanced hotel-level data set included 85,939 observations across 3,996 hotels.

To augment our data set, we sought supplementary information from external sources. First, we searched the hotel location on [map.baidu.com](#) to acquire latitude and longitude coordinates for each hotel. We then used the coordinates to calculate the geographic distances between hotels. Subsequently, we accessed [Anjuke.com](#)—a leading real estate application in China—to obtain the historical commercial real estate and leasing prices for each postal code area. These variables were used as the exogenous instrumental variables for causal identification.

4.2. Variables

4.2.1. Dependent Variable: Self-Injecting Intensity.

Although anyone can leave a review on [Qunar.com](#), [Ctrip.com](#) only allows consumers to post reviews, leading to observable differences in the distribution of positive ratings between the two platforms (Mayzlin et al. 2014, Zhuang et al. 2018, Kokkodis et al. 2022). To rule out the possibility of disparate customer preferences, we conducted a comparative analysis of review statistics for the same hotels on [Qunar.com](#) and [Ctrip.com](#). The results revealed that both the ratings and review volumes of the two websites are strongly correlated ($\gamma_{rating} = 0.873$, $p = 0.000$; $\gamma_{volume} = 0.902$, $p =$

0.000). Therefore, customer preferences did not differ significantly across the two websites. Further, Zhuang et al. (2018) suggested that neutral reviews (i.e., three-star and four-star ratings) should be intact on both the “open” and “close” platforms. We used the difference in the ratio of positive reviews to both neutral and positive reviews between [Qunar.com](#) and [Ctrip.com](#) to represent the intensity of the hotel’s self-injecting behavior. For hotel i at time t , we measured the self-injecting intensity using the following equation:

$$Inject_{i,t} = \frac{5StarReviews_{i,t}^{Qunar}}{3 + 4 + 5StarReviews_{i,t}^{Qunar}} - \frac{5StarReviews_{i,t}^{Ctrip}}{3 + 4 + 5StarReviews_{i,t}^{Ctrip}}. \quad (1)$$

4.2.2. Independent Variables. We defined competition as a situation in which hotels vie for the same customer groups (Forman et al. 2009, Gutt et al. 2019). Then, we identified the competitors of a hotel as the hotels located in the same geographical region (i.e., postal code) as the focal hotel (Liu et al. 2018, Alyakoob et al. 2021). Rival refers to evenly matched competitors with heightened psychological stakes, historical interaction, and mutual recognition and respect (Kilduff et al. 2016, Pike et al. 2018). We identified rivals using both objective filtration and subjective evaluation to conform to its theoretical definition as much as possible. We first filtered out rival candidates for each hotel using objective indices proposed by Kilduff et al. (2016). We then telephone interviewed the managers of all the hotels in our sample, asking them to provide a subjective evaluation and select rivals from the identified candidates. Online Appendix 1 shows the specific procedures for rival identification. We observe significant differences in the values of rivals with and without subjective measures, as shown in Table 1. These differences are evident in their mean, standard deviation, and value range. The results highlight substantial variations in how hotel managers select rivals when using objective indexes to identify rival candidates.⁶

4.2.3. Control Variables. We included several covariates for a more accurate test of the hypotheses. We controlled for a set of review-related variables, such as *cumulative review count*, *average rating*, *number of negative reviews*, and *number of managerial responses*, as they have

Table 1. Statistics of Rivals with and Without Subjective Measures

Rival	Mean	Standard deviation	Minimum	Median	Maximum
With subjective measure	2.051	1.643	0	2	7
Without subjective measure	2.738	4.202	0	1	34

Table 2. Operationalization of Variables

Variable	Abbreviation	Operationalization	Sources
Self-injecting intensity	$Inject_{i,t}$	The difference in the ratio of positive reviews between Qunar.com and Ctrip.com for a given hotel i at time period t .	Qunar.com and Ctrip.com
Number of nearby competitors	$Compe_{i,t}$	The number of hotels that are located in the same region (same postal code) with the focal hotel i at time period t .	Ctrip.com
Number of nearby rivals	$Rival_{i,t}$	The number of identifiable, historical, and evenly matched opponents with heightened psychological stakes and recognition for hotel i at time period t .	Ctrip.com and Telephone Interview
Self-injecting intensity of rivals	$Rival_Inject_{i,t}$	The average self-injecting intensity of rivals for hotel i at time t .	Qunar.com and Ctrip.com
Self-injecting intensity of nonrival competitors	$NRival_Inject_{i,t}$	The average self-injecting intensity of nonrival competitors for hotel i at time t .	Qunar.com and Ctrip.com
Chain affiliation	$Chain_{i,t}$	Dummy variable, equal to 1 if the hotel i is an affiliate to a brand chain, at time t .	Ctrip.com
Cumulative review volume	$RevVol_{i,t}$	The number of cumulative consumer reviews for hotel i at time t .	Ctrip.com
Average rating	$Rating_{i,t}$	The average rating of hotel i at time t .	Ctrip.com
Number of negative reviews	$NegRev_{i,t}$	The number of 1-star and 2-star reviews that hotel i received at time t .	Ctrip.com
Number of negative reviews received by the rivals of a given hotel	$RNegRev_{i,t}$	The number of 1-star and 2-star reviews that the rivals of hotel i received at time t .	Ctrip.com
Number of managerial responses	$Resp_{i,t}$	The number of managerial responses to consumer reviews for hotel i at time t .	Ctrip.com
Hotel age	$Age_{i,t}$	The number of months that has elapsed for hotel i at time t since the hotel was established.	Ctrip.com
Hotel size	$Size_{i,t}$	Number of rooms for hotel i at time t .	Ctrip.com
Hotel star ^a	$Star_{i,t}$	Star evaluation (from 1-star to 5-star) by Ministry of Culture and Tourism of China for hotel i at time t .	www.zwfw.mct.gov.cn/hotel
Business service	$BusiServ_{i,t}$	Dummy, equal to 1 if hotel i has a business center or conference room at time t .	Ctrip.com
Fitness	$Fitness_{i,t}$	Dummy, equal to 1 if hotel i has a fitness or gym at time t .	Ctrip.com
Restaurant	$Resta_{i,t}$	Dummy, equal to 1 if hotel i has a restaurant at time t .	Ctrip.com
Swimming pool	$Swim_{i,t}$	Dummy, equal to 1 if hotel i has a swimming pool at time t .	Ctrip.com
Parking	$Park_{i,t}$	Dummy, equal to 1 if hotel i has a private parking lot at time t .	Ctrip.com
Airport transport service	$AirTrans_{i,t}$	Dummy, equal to 1 if hotel i provides an airport transport service at time t .	Ctrip.com
Real estate price	$Price_{k,t}$	The commercial property prices for postal code k at time t . Measured as: ¥/m ² .	www.anjoke.com
Real estate rental	$Rental_{k,t}$	The commercial property rental for postal code k at time t . Measured as: ¥/m ² /month.	www.anjoke.com
Geographic distance	$Dist_{i,j}$	The geographic distance (in kilometers) between hotel i and hotel j .	www.map.baidu.com

Note. i , hotels; k , postal code area; t , time window.

^aAccording to the “Star Hotel Evaluation Standard of the People’s Republic of China,” the hotel is divided into five standards from one-star to five-stars. The higher the star level, the more luxurious the hotel.

been found to predict self-injecting behaviors in previous studies (Hu et al. 2012, Mayzlin et al. 2014, Luca and Zervas 2016, Kumar et al. 2018b). We also controlled for hotel-specific factors, such as *chain affiliation*, *age*, *star*, *size*, and multiple dummies for hotel services. Because hotels may improve their services by building a parking lot or opening up a restaurant, all hotel-

specific variables were constructed in a time-variant manner, meaning that they could vary between months. Table 2 presents the operationalizations of the dependent variable, independent variables, and control variables. Table A.1 and Table A.2 in Online Appendix 2 show the descriptive statistics of variables and correlations among variables.

4.3. Model Specification

We constructed variables at the hotel-month level. The two independent variables, $Rivals_{i,t}$ and $Rival_Inject_{i,t}$, are estimated in Equation (2), where α_0 is the model intercepts, γ_i captures unobserved hotel-specific heterogeneity (hotel-level fixed effects), η_t captures temporal-specific heterogeneity (time-level fixed effects), and ε_{ij} indicates the residual term; $X_{i,t}$ is a vector of variables that controls hotel-specific characteristics, such as the hotel's official star categorization (one to five stars), hotel age, chain affiliation (chain or nonchain), and hotel services dummies (restaurant, business service, parking, swimming, airport transport service, and fitting service); and $Rev_{i,t}$ is a vector of variables that control review-specific characteristics, such as the hotel's self-injecting intensity in the last month, cumulative review count, the average rating of a hotel, and the number of negative reviews for hotel i received at time t . The key parameters of interest in these models are α_1 and α_2 . The valance and the significance level of α_1 indicate whether there is a positive or negative relationship between hotel's self-injecting intensity and the number of rivals nearby (Hypothesis 1a and Hypothesis 1b). The coefficient, α_2 , determines whether the self-injecting intensity of a hotel increases the self-injecting intensity of its rivals (Hypothesis 2a). The variable $NRival_Inject_{i,t}$ was added to Equation (3). Hypothesis 2b suggests that a hotel's self-injecting intensity is more responsive to the self-injecting behavior of its rivals than to the self-injecting behavior of its nonrival competitors. To test Hypothesis 2b, we estimated the coefficients of $Rival_Inject_{i,t}$ and $NRival_Inject_{i,t}$ in a standardized regression model (Equation (3)) and compared the difference between the two estimated coefficients ($\beta_2 - \beta_3$) using the Wald test (Chou and Bentler 1990). For Hypothesis 3a and Hypothesis 3b to be held true, in Equation (4), ζ_3 should be positive, whereas ζ_4 should be negative. Our ordinary least squares (OLS) specification can be summarized as follows:

$$Inject_{i,t} = \alpha_0 + \alpha_1 \cdot Rivals_{i,t} + \alpha_2 \cdot Rival_Inject_{i,t} + \alpha_3 \cdot X_{i,t} + \alpha_4 \cdot Rev_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}, \quad (2)$$

$$Inject_{i,t} = \beta_0 + \beta_1 \cdot Rivals_{i,t} + \beta_2 \cdot Rival_Inject_{i,t} + \beta_3 \cdot NRival_Inject_{i,t} + \beta_4 \cdot X_{i,t} + \beta_5 \cdot Rev_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}, \quad (3)$$

$$Inject_{i,t} = \zeta_0 + \zeta_1 \cdot Rivals_{i,t} + \zeta_2 \cdot Rival_Inject_{i,t} + \zeta_3 \cdot Rivals_{i,t} \cdot Reputation_{i,t} + \zeta_4 \cdot Rival_Inject_{i,t} \cdot Reputation_{i,t} + \zeta_5 \cdot X_{i,t} + \zeta_6 \cdot Rev_{i,t} + \gamma_i + \eta_t + \varepsilon_{i,t}. \quad (4)$$

4.4. Endogeneity and Identification

Despite the inclusion of control variables and different fixed effects, it is plausible that our OLS estimates of α_1

and α_2 are skewed, because other omitted variables are likely to exist for which we cannot explicitly control. Deterrence theory suggests that a firm's decision to engage in unethical behaviors is determined by the possibility and severity of the sanctions for those unethical behaviors (Trinkle et al. 2021). The risks of self-injecting behavior being caught and the severity of the sanctions for self-injecting may influence a firm's self-injecting intensity. These factors, however, are unobserved. To account for the endogeneity generated from omitted variables, we developed an instrumental variable (IV) identification technique. Any potential instrument, Z , for our endogenous variables, $Rivals$ and $Rival_Inject$, must satisfy two conditions. First, it must be sufficiently correlated with our endogenous variables—that is, it should significantly affect $Rivals$ and $Rival_Inject$. Second, the IV must satisfy the exclusion restriction; it must affect self-injecting intensity exclusively via its influence on $Rivals$ and $Rival_Inject$. Importantly, the instrument requires that $\text{cov}(Rivals_{i,t}, Z) \neq 0$, $\text{cov}(Rival_Prom_{i,t}, Z) \neq 0$, and $\text{cov}(Z, \varepsilon_{i,t}) = 0$.

We identified the $Price_{i,t}$ and $Rental_{i,t}$ of commercial real estate in the area where the focal hotel was located⁷ as the instrumentals of $Rivals_{i,t}$. Because hotels generally have the option of buying or renting their property, it seems natural that the decision to establish a rival hotel in an area would be significantly correlated with the $Price_{i,t}$ and $Rental_{i,t}$ in that area. The high and statistically significant correlations in these connections, as shown in Table A.2, corroborate this. Using an OLS regression approach, we were able to reject the null hypothesis that there is no relationship between $Rental_{i,t}$ and $Rivals_{i,t}$ ($b = 0.938$, $t = 12.87$) at the 0.1% level. For the rental increase by one unit (¥1/m²/month), the number of nearby rivals decreased by 0.25. In addition, after controlling $Rivals_{i,t}$, neither $Price_{i,t}$ nor $Rental_{i,t}$ predicts the self-injecting intensity of a hotel.⁸ We used the rival hotel's consumer review response rate ($R_Resp_{i,t}$) as the instrumental for $Rival_Inject_{i,t}$. Kumar et al. (2018a) posit that there are two options for firms to intervene in online word-of-mouth: either online review manipulation or online managerial responses. For a given hotel, the resources for review response would be negatively correlated to that of review manipulation. Considering the constraints on resources, when a hotel allocates more resources to responding to consumer reviews, it will likely have fewer resources available for engaging in review manipulation. An OLS regression supports the negative relationship between $R_Resp_{i,t}$ and $Rival_Inject_{i,t}$ ($b = -0.084$, $t = -23.34$). Managerial response is used to show customer care to consumers who have already posted a rating (Chen et al. 2019). A focal hotel's managerial responses would not threaten its rival's ranking. Therefore, a focal hotel's self-injecting intensity does not respond to the review response rate of its rivals. Moreover, we utilize the method proposed

by Lewbel (2012), constructing orthogonal instruments mathematically from variables specified as exogenous. This approach has found successful application across various disciplines, such as economics, marketing, operations management, and information systems (Kashyap and Murtha 2017, Hong et al. 2018, Heim et al. 2022, Wang et al. 2022b). The constructed equations are delineated as follows:

$$Y_1 = \varphi_1 X' + \varphi_2 Y_2 + \varepsilon_1, \quad (5)$$

$$Y_2 = \theta X' + \varepsilon_2. \quad (6)$$

Here, Y_1 represents self-injecting intensity; Y_2 denotes endogenous variables, that is, $Rival_{i,t}$ and $Rival_Inject_{i,t}$, and X' encompasses a set of exogenous variables. The estimation involves two stages. In the first stage, the regression of Equation (6) is performed to derive the residual estimate $\hat{\varepsilon}_2$ and construct $(Z - \bar{Z})\hat{\varepsilon}_2$ as an instrumental variable. Here, the exogenous variable vector Z is a subset of X or $Z = X$ and $\bar{Z}$ is the mean value of Z . In our model, we treat *Real Estate Rentals* and *Price* in postal code, and *Number of Hotels* in postal code as exogenous, and construct instruments for the endogenous terms, including $Rival_{i,t}$ and $Rival_Inject_{i,t}$. In the second stage, the effect of $Rival_{i,t}$ and $Rival_Inject_{i,t}$ on $Inject_{i,t}$ is obtained using the constructed instrumental variable, which is introduced into Equation (5). The rationale behind this approach lies in the fact that the generated instruments (Z_j) can be constructed from the product of the mean-centered form of those exogenous variables (X_j) and the residuals obtained from the “first-stage” regression: $Z_j = (Z - \bar{Z})\hat{\varepsilon}_2$, where $\hat{\varepsilon}_2$ is the vector of residuals from the “first stage” regression of each endogenous regressor on all exogenous regressors, including a constant vector. Lewbel (2012) shows that Z_j are valid instruments under the assumption that $cov(X_j, \rho \cdot \varepsilon) = 0$. Finally, following Imbert et al. (2022), we construct a Bartik instrument for *Rival* (*Rival_Inject*) using the original value of *Rival* (*Rival_Inject*) times the growth rate of the *Rivals* (*Rival_Inject*) at the regional postal code level. The constructed Bartik instrument is highly correlated with *Rival* (*Rival_Inject*). Studies have shown that the Bartik instrument would be exogenous after fixing hotel and time effects (Goldsmith-Pinkham et al. 2020, Forbes and Kosova 2022).

5. Estimation Results

5.1. OLS Estimates

Table 3 shows the results of the OLS estimates, and Table 4 demonstrates the results of IV regressions. To avoid multicollinearity, the variables in Tables 3 and 4 are mean-centered before creating the interaction terms. We report the average variance inflation factors (VIF) for each model in Table 3. The VIFs for these models were far below the threshold of five (O'Brien 2007), suggesting that the multicollinearity issue was not a

concern in this study. We lagged the key independent variables by one month. We first estimated a set of two-way fixed-effect (fixed hotel and month) models (Models 1–5). Model 1 is the baseline model that only includes a set of control variables. Model 2 shows that the number of nearby rivals negatively predicts hotel's self-injecting intensity ($\beta = -0.032, p < 0.010$), thereby supporting Hypothesis 1b. Holding the control variables constant, the number of nearby rivals rises from one to two leading to a 3.2% decrease in hotel's self-injecting intensity. Hypothesis 2a was supported as the coefficient of *Rival_Inject* in Model 2 is positive and significant ($\beta = 0.097, p < 0.001$). The self-injecting intensity of nonrival competitors was added to Model 3. Model 3 shows that a focal hotel's self-injecting intensity increases by 0.095% for every 1% rise in that of its rivals and increases by 0.041% for every 1% rise in that of its nonrival competitors. These results confirmed Hypothesis 2b, which argues that hotel's self-injecting intensity is more responsive to the self-injecting behavior of their rivals than to those of their nonrival competitors. To formally compare the effects of rival and nonrival injecting, Model 4 employed a standardized regression. Although both rival injecting ($\beta = 0.051, p < 0.001$) and nonrival injecting ($\beta = 0.022, p < 0.001$) fosters self-injecting intensity, the coefficient of rival injecting was greater than that of nonrival injecting. The results of the Wald test show that the difference in the two coefficients was statistically significant ($F(1, 3225) = 14.10, p < 0.001$), supporting Hypothesis 2b. To test Hypothesis 3a and Hypothesis 3b, Model 5 included two interaction terms: $Rival_{i,t} \times Reput_{i,t}$ and $Rival_Inject_{i,t} \times Reput_{i,t}$. Reputation is an abstract concept that cannot be measured directly. Previous studies used multiple objective indexes to proxy for hotel reputation (Yang et al. 2021, Filippas et al. 2022). Here, we used hotel stars to measure hotel reputation. In the robust checks, we use hotel rating and chain affiliation as the alternative measurements of hotel reputation (see Section 5.3). Model 5 shows that $Rival_{i,t}$ negatively predicted self-injecting intensity ($\beta = -0.019, p < 0.050$), whereas the coefficient of $Rival_{i,t} \times Reput_{i,t}$ was found to be negative and significant ($\beta = -0.012, p < 0.001$). Therefore, hotel reputation strengthens the negative relationship between the number of nearby rivals and online self-injecting intensity, supporting Hypothesis 3a. Hypothesis 3b is supported in the same way as $Rival_Inject_{i,t}$ positively predicted self-injecting ($\beta = 0.082, p < 0.001$), whereas the coefficient of $Rival_Inject_{i,t} \times Reput_{i,t}$ was found to be negative and significant ($\beta = -0.034, p < 0.010$). We also estimated a one-way fixed-effect model (Models 6–9). The one-way fixed model fixes the hotel effect. The results presented in Models 6–9 are identical to those presented in Models 2–5, providing additional support for our hypotheses.

Table 3. Results of OLS Regressions on Self-Injecting Intensity

Variable	Model 1	Model 2	Model 3	Model 4 standardized	Model 5	Model 6	Model 7	Model 8 standardized	Model 9
$\ln_Rival_{i,t}$		−0.032** (−3.183)	−0.031** (−3.166)	−0.100** (−3.190)	−0.019* (−2.337)	−0.037*** (−3.864)	−0.037*** (−3.858)	−0.147*** (−5.429)	−0.024* (−2.102)
$Rival_Inject_{i,t}$		0.097*** (8.903)	0.095*** (8.753)	0.051*** (8.752)	0.082*** (13.27)	0.093*** (8.549)	0.090*** (8.363)	0.048*** (8.268)	0.076*** (6.028)
$NRival_Inject_{i,t}$			0.041*** (4.725)	0.022*** (4.731)			0.043*** (4.962)	0.023*** (4.855)	
$\ln_Rival_{i,t} \times Reput_{i,t}$					−0.012*** (−4.878)				−0.013*** (−4.036)
$Rival_Inject_{i,t} \times Reput_{i,t}$					−0.034** (−2.600)				−0.038† (−1.661)
$Rating_{i,t}$	0.011** (3.168)	0.010** (2.658)	0.006 (1.530)	0.020** (2.590)	0.009*** (4.399)	0.003 (0.861)	0.004 (0.874)	0.006 (0.695)	0.003 (0.820)
$\ln_RevVol_{i,t}$	−0.110*** (−43.61)	−0.110*** (−43.53)	−0.109*** (−38.28)	−0.743*** (−42.74)	−0.110*** (−79.58)	−0.098*** (−36.12)	−0.098*** (−36.10)	−0.639*** (−35.85)	−0.098*** (−36.22)
$\ln_Resp_{i,t}$	0.018*** (11.03)	0.018*** (11.01)	0.018*** (9.655)	0.040*** (7.888)	0.018*** (14.961)	0.016*** (8.697)	0.016*** (8.725)	0.121*** (8.912)	0.016*** (8.795)
$\ln_NegRev_{i,t}$	0.088*** (7.908)	0.088*** (7.926)	0.094*** (7.599)	0.145*** (11.12)	0.088*** (12.00)	0.109*** (8.888)	0.109*** (8.912)	0.049*** (8.920)	0.109*** (8.928)
$Age_{i,t}$	0.001 (1.236)	0.001 (1.177)	0.000*** (5.227)	0.001 (0.933)	0.001 (0.873)	−0.000** (−2.875)	−0.000** (−3.010)	−0.000 (−1.322)	−0.000** (−3.043)
Constant	0.250*** (14.13)	0.282*** (14.30)	0.175*** (13.39)	0.062** (3.223)	0.271*** (18.62)	0.196*** (16.95)	0.196*** (16.95)	0.070*** (7.006)	0.183*** (13.53)
Fixed effect: Hotel	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fixed effect: Month	Yes	Yes	Yes	Yes	Yes	No	No	No	No
Observations	85,726	70,062	70,062	70,062	70,062	70,062	70,062	70,062	70,062
No. of hotels	3,996	3,226	3,226	3,226	3,226	3,226	3,226	3,226	3,226
Average VIF	1.36	1.73	1.72	1.37	1.73	2.72	2.53	1.92	2.49
Within R^2	0.134	0.137	0.138	0.138	0.138	0.128	0.128	0.128	0.128
Between R^2	0.216	0.213	0.214	0.226	0.218	0.216	0.216	0.214	0.220
Overall R^2	0.128	0.126	0.126	0.131	0.125	0.126	0.126	0.127	0.124

Notes. Hotel dummies are $Chain_{i,t}$, $BusiServ_{i,t}$, $Fitness_{i,t}$, $Resta_{i,t}$, $Park_{i,t}$, and $AirTrans_{i,t}$. Models 4 and 8 are standardized regressions. t statistics are shown in parentheses. VIF, variance inflation factor.

*** $p < 0.001$; ** $p < 0.010$; * $p < 0.050$; † $p < 0.100$.

5.2. Instrumental Variable Regression

We further moved to our IV estimates that are reported in Table 4. For each independent variable, $Rival$ or $Rival_Inject$, we have three instrumental variables. The first stage Kleibergen-Paaprk Wald F statistics for $Rival$ and

$Rival_Inject$ were 316.3, 524.7, and 4,564, respectively, indicating that our instruments were sufficiently powered (de Vaan et al. 2021). Columns (1)–(3) in Table 4 estimate the influence of $Rival$ and $Rival_Inject$ by adding the three instruments one by one. Column (4) in

Table 4. Results of 2SLS Regressions

	(1)	(2)	(3)	(4)
$\ln_Rival_{i,t}$	−1.056* (−2.050)	−0.076*** (−4.065)	−0.047** (−2.577)	−0.063*** (−4.212)
$Rival_Inject_{i,t}$	0.088* (2.101)	0.156*** (10.76)	0.158*** (9.451)	0.168*** (19.67)
Control variables	Yes	Yes	Yes	Yes
IV1: Rental/R_Resp	Yes	No	No	Yes
IV2: Lewbel Instrument	No	Yes	No	Yes
IV3: Bartik Instrument	No	No	Yes	Yes
Fixed effect: hotel	Yes	Yes	Yes	Yes
Fixed effect: month	Yes	Yes	Yes	Yes
Observations	66,241	70,062	70,016	66,241
No. of hotels	3,161	3,226	3,180	3,161
Average VIF	3.880	1.470	1.820	1.530
Kleibergen-Paaprk Wald F	316.3	524.7	4,564	508.1

Notes. The control variables are the same as they are in Table 3. To save space, the coefficients of the control variables and the results of the first stage regressions are omitted. They are available upon request.

*** $p < 0.001$; ** $p < 0.010$; * $p < 0.050$; † $p < 0.100$.

Table 4 estimates the influence of *Rival* and *Rival_Inject* by adding the three instruments once at a time. The Two Stage Least Squares (2SLS) estimates were broadly consistent with our OLS estimates in the sense that they showed a robust negative effect of *Rival_{i,t}* and a positive effect of *Rival_Inject_{i,t}* on hotel's self-injecting intensity.

5.3. Robustness Checks

We performed a series of additional tests to fortify our findings (see Online Appendix 3). First, we split the sample into the high and low rating, as well as chain and nonchain hotel groups and executed the models in these four groups independently (see Tables A.3 and A.4). In the main analysis, we use hotel star to measure hotel reputation and create interaction terms to test Hypothesis 3a and Hypothesis 3b. In this section, we establish the robustness of our results using alternative measures (hotel's average rating and chain affiliation) of hotel reputation and tested Hypothesis 3a and Hypothesis 3b using between-group comparisons (see Online Appendix 3.1). Second, we used the maximum self-injecting intensity of nearby rivals as an alternative measure of the independent variable, *Rival_Inject_{i,t}* and re-estimated the models (see Table A.5). The underlying logic of this measurement can be justified by the winner-takes-all phenomenon in the platform economy (Wang et al. 2022c, 2023). Third, our additional tests ruled out alternative explanations for the relationship between rivalry and self-injecting intensity (see Table A.6 and Table A.7). Fourth, we re-estimated the models by using alternative observation windows for both the dependent and independent variables (see Table A.8). The results of these additional tests were largely identical to those of the main analysis, further ensuring the validity of the findings. Fifth, we divide our sample into two groups, those with rivalry and those without rivalry, and perform a subsample analysis (see Table A.9). Sixth, we investigate two ways of review manipulation: self-injecting favorable reviews and posting unfavorable reviews about others. Here, we examine how rivalry influences the behavior of injecting unfavorable reviews toward rivals (see Table A.10). Lastly, we performed a follow-up phone interview. The follow-up phone interview yielded a data set consisting of 335 hotels and 7,126 observations. We then re-estimated our models, this time controlling for the demographics of the general managers (see Table A.11). The results presented in Table A.11 closely mirrored those in Tables 3 and 4, thereby demonstrating the robustness of our findings.

6. Discussion and Implications

6.1. Discussion and Key Findings

Our study aimed to answer a set of research questions with progressive relationships: Does rivalry fuel or impede online self-injecting behaviors? How does the

effect of rivalry vary with hotel heterogeneity? To answer these questions, we develop a set of theoretically driven hypotheses and rigorously test them by leveraging an exhaustive data set from multiple sources. Contrary to previous studies that assert that competition fuels unethical behaviors, we discovered robust evidence that rivalry inhibits, rather than fuels, online self-injecting behavior (supporting Hypothesis 1b). This striking finding survived numerous robustness tests, including a set of subsample analyses, the use of alternative measures of dependent and instrumental variables, the use of alternative observation time windows, and several endogeneity tests. This seemingly counterintuitive conclusion suggests that rivalry-induced deterrence outweighs the gains from risk-taking behaviors. The new finding proves the necessity of differentiating rivalry from competition and underscores that rivalry and competition could play distinct roles in predicting unethical behaviors.

We further uncovered a causal relationship between a hotel's self-injecting intensity and its rival's self-injecting intensity, thus supporting Hypothesis 2a. This finding extended the prevalent peer influence phenomenon from ethical to unethical behaviors. The underlying mechanisms of peer influence on ethical behaviors might differ from those on unethical behaviors. For ethical behaviors, the influence of peers typically manifests through a variety of mechanisms, such as observational learning, social norms, imitation, and homophily (Bapna and Umyarov 2015, Beer et al. 2021, Li et al. 2021, Cui et al. 2024). A more compelling explanation for "peer influence" in the context of unethical behaviors (self-injecting favorable reviews) is to minimize losses. The ranking of a focal hotel is negatively correlated with that of its rivals. Seeing rival self-injecting favorable reviews, hotels may engage in self-injecting to offset the disadvantages caused by the self-injecting behaviors of rivals.

We confirmed that a hotel's online self-injecting intensity is more responsive to the self-injecting behavior of nearby rivals than to that of nearby nonrival competitors (supporting Hypothesis 2b). We show that self-injecting intensity increases by 0.095% for every 1% rise in that of its rivals, which is more than twice the corresponding increase observed in that of its nonrival competitors (0.041%). The result implies that observing a rival's self-injecting behavior causes the focal hotel to self-inject in a more intensive way as a retaliatory response than observing nonrival's self-injecting behaviors. Although previous studies suggested that rivalry fosters greater motivations and efforts than nonrival competition (Kilduff 2014, Pike et al. 2018), our findings present a nuanced perspective by showing rivalry is a "double-edged sword." The confirmation of these hypotheses suggests that unethical behaviors are positive responses to the unethical behaviors of rivals but exhibit a negative response to the mere presence of rivals.

As expected, we uncovered divergent moderating roles of reputation on the relationship between rival and self-injecting intensity. We show that hotel reputation plays a crucial role in strengthening the negative relationship between the number of rivals and self-injecting behaviors (supporting Hypothesis 3a). Economic literature has proved that collusion, collaboration, and contracts are more likely to be formed among reputable firms than less reputable firms due to a high level of mutual trust (Cai and Obara 2009). Our results affirm this line of work by showing that mutual forbearance, functioning as a tacit form of collusion, is more prone to be established among reputable rival hotels than less reputable counterparts. We also found that hotel reputation weakens the positive relationship between rival injecting and self-injecting (Hypothesis 3b). This finding corroborates the important role of firm reputation in alleviating unethical behaviors (Hollenbeck 2018, Kokkodis and Lappas 2020, Yang et al. 2021).

Taken together, our empirical findings unveil three compelling insights that have profound theoretical and practical implications for platforms, marketers, and consumers. First, we find that rivalry stimulates the formation of mutual forbearance in acting unethically. Second, the delicate balance of mutual forbearance collapses when one party starts to unilaterally self-inject, triggering retaliatory self-injection from the other party. Third, we establish that the reputation of a firm emerges as a pivotal factor, strengthening the formation of mutual forbearance and weakening its collapse concurrently. In the following sections, we will delve into these results to unravel their implications for research and practice.

6.2. Theoretical Contributions

This study advances theory in several significant ways. First, this study enriches the literature on online review manipulation by identifying rivalry as a hitherto undiscovered inhibitor of online self-injecting behaviors, thus helping scholars better understand when and why firms behave unethically and offering insights on how to curtail such unethical behaviors. Previous studies have generally concurred that competition fosters self-injecting behaviors (Mayzlin et al. 2014, Luca and Zervas 2016, Gutt et al. 2019, Park et al. 2021). However, these studies often treated rivalry and nonrival competition interchangeably, leaving us starkly uninformed about the specific relationship between rivalry and a firm's self-injecting behaviors. Diverging from the previous studies situated in the context of competition, this study is grounded in rivalry, which is a related but theoretically different concept. By conceptually differentiating rivalry from nonrival competition, we provide concrete evidence that rivalry and nonrival competition play distinct roles in predicting a firm's self-injecting behaviors. Drawing on the collective regulatory focus framework

and deterrence theory, we show that rivalry-induced deterrence of acting unethically shifts review management team's mindset from a promotion focus (maximize the gains of self-injection) to a prevention focus (minimize the risks/costs of self-injection) that dampens hotel's self-injecting intensity. Our discovery of the negative relationship between rivalry and self-injecting intensity challenges the long-held belief in previous competition literature and opens up a new direction of review manipulation governance: the cultivation of transparent rivalry. Thus, our work further extends the existing paradigm on review manipulation and enriches the understanding of the competitive determinants influencing online review manipulation. Further, this study provides an opportunity to quantify the effect of rivalry on unethical behaviors such as self-injecting intensity studied in this paper, thereby offering a valuable contribution to the empirical exploration of these phenomena.

Second, this study substantially extends the literature on self-injecting behaviors by investigating how the number of rivals can affect self-injecting behaviors: an unpretentious yet important contextual attribute of rivalry that has received surprisingly limited attention in previous research. Extant studies on competitive contexts and unethical behaviors have primarily focused on the effects of the individual attributes of competitors (Tzini and Jain 2018), the relationship between competitors (Kilduff et al. 2016), and the competitive intensity between competitors (Yip et al. 2018, Feltovich 2019). The number of rivals or competitors is often a readily accessible cue in competitive contexts. For example, e-commerce platforms explicitly display the number of available sellers under certain search criteria. However, the effect of rivalry pool size in predicting self-injecting intensity has yet to be examined in previous literature. This study addresses this notable gap by showing that self-injecting intensity is indeed responsive to the presence of nearby rivals and that an increase in the number of nearby rivals reduces hotel's self-injecting intensity. In doing so, our research sheds light on an unexplored facet of rivalry dynamics, offering valuable insights into how the sheer number of rivals can shape and influence self-injecting behaviors in competitive contexts.

Third, in contrast to prevailing literature that generally holds a unitary perspective on rivalry (Ager et al. 2022, Sharapov and Ross 2023, Akerlof et al. 2024), this study extends the rivalry discourse by uncovering the multifaceted nature of rivalry. Although past research has criticized the role of rivalry in stimulating unethical behaviors (Kilduff et al. 2016, Ager et al. 2022), our conceptual exploration of the rise and fall of mutual forbearance in rivalry offers a dynamic viewport, revealing the double-edged sword effect of rivalry. We contribute by discovering that rivalry-induced deterrence leads to the formation of mutual forbearance in

acting unethically, which ultimately attenuates self-injecting intensity. We also show that hotels will retaliate against their rivals in a more aggressive way than against nonrival competitors once the tacit mutual forbearance crumbles. These results underscore that rivalry incorporates not only confrontation but also collaboration and retaliation. Understanding the multifaceted nature of rivalry delivers great theoretical values to inform policymakers to fully utilize rivalry to reduce online review manipulations. We further advance this stream of research by proposing hotel reputation as an effective way to foster the advantages and circumvent the disadvantages of rivalry. This initial attempt has opened up a strategic window for future studies to explore other catalytic factors of rivalry.

Fourth, we enrich the literature by extending the prevalent “peer influence” phenomenon beyond ethical behaviors to encompass unethical behaviors. More importantly, we enrich theory by providing the collapse of rivalry-induced deterrence as an alternative mechanism for peer influence in unethical behaviors. Regarding ethical behaviors, the influence of peers takes effect through a variety of mechanisms, such as observational learning, social norms, imitation, and homophily (Bapna and Umyarov 2015, Li and Wu 2018, Beer et al. 2021, Li et al. 2021, Wang et al. 2022a). These mechanisms may not be directly applicable to unethical behaviors. Our study proposes a more compelling explanation for the phenomenon of peer influence in self-injecting behaviors by emphasizing the collapse of rivalry-induced deterrence. Analogous to the old saying “A tainted person is less likely to expose another tainted person,” when hotels witness their rivals engaging in self-injecting favorable reviews, the deterrence associated with the expectation that those rivals would expose such unethical behaviors diminishes significantly. Consequently, hotels become more inclined to follow the unethical behavior of their rivals as the rivalry-induced deterrence collapses. By presenting this viewpoint, we gain a deeper comprehension of the processes that propel peer influence in the realm of unethical actions, thereby providing significant contributions to the knowledge of how online self-injecting behaviors are impacted by peer actions.

6.3. Managerial Implications

Our findings suggest a range of practical implications for platforms, sellers, and consumers. We found that rivalry-induced deterrence is a critical force that inhibits hotel’s self-injecting behaviors. It is imperative for platforms to emphasize the rivalry-induced deterrence of self-injecting behaviors. Platforms, as key stakeholders, are advised to upgrade their fake review filter systems regularly to detect self-injecting behaviors swiftly and publicize hotels that act unethically. Then, platforms are suggested to increase the possibility,

severity, and promptness of the punishment for hotel’s self-injecting behaviors. Meanwhile, hotels that do not engage in self-injecting behaviors should be recognized and rewarded. Moreover, platforms can foster a culture of mutual regulation among hotels and encourage whistle-blowing mechanisms for reporting unethical behaviors. The establishment of transparent channels for reporting such behaviors is essential. Given the apparent role of rivalry in reducing self-injecting behaviors, platforms should strategically consider how system designs and policies can cultivate firms’ feelings of rivalry and avoid nonrival competition. For instance, platforms may aim to design hotel reservation systems to encourage the antecedents of rivalry, such as fostering repeated offline interactions and promoting peer recognition among hotels. In the current online review systems, reservation platforms recommend hotels based on the evaluations of their previous consumers. We strongly recommend platforms consider including evaluations from peer hotels as an alternative index for recommending hotels to potential consumers. The ranking of hotels on the reservation platforms could be determined by the ratings of both previous consumers and peer hotels. Introducing peer evaluations to the current online review systems could attenuate the bias of consumer ratings and facilitate the formation of rivalry, which hinders self-injecting behaviors.

Hotels on reservation platforms are suggested to foster mutual forbearance and enable long-standing reciprocal relationships between rivals to persist. Hotels are also encouraged to scrutinize the behaviors of their rivals regularly to maintain a healthy competitive environment. Our research reveals that the collapse of mutual forbearance among rivals can lead to retaliatory self-injecting behaviors of the other party, resulting in a lose-lose situation. Instead, we strongly suggest hotels remain rational and address rival’s unethical behaviors by reporting them to platforms and relevant authorities. Hotel managers and owners that face intense rival competition must strengthen the monitoring and governance mechanisms to ensure their legal compliance if they want to mitigate the risk of government sanctions and reputation damage. It is important for hotel managers to strike a balance, avoiding an excessive focus on outperforming rivals that may prioritize profits over ethical business practices. Hotel executives may want to take explicit measures to prevent rival’s unethical behaviors from influencing their moral judgments and strategic decisions, such as desensitizing data from rivals and relying on objective decision criteria.

Our findings indicate that hotels tend to engage in self-injecting behaviors when their rivals have done the same. When consumers become aware of a focal hotel self-injecting favorable reviews, they normally turn to the rivals of the focal hotel. In such instances, consumers are suggested to discount or at least be skeptical

about the credibility of the favorable online reviews of rivals. This cautious approach enhances consumer discernment and encourages a more informed decision-making process when evaluating hotels in a competitive online landscape.

7. Concluding Remarks and Future Directions

7.1. Limitations and Future Research

This study has several limitations that provide fruitful avenues for future research. First, this study focuses on the effect of rivalry on self-injecting favorable online reviews. However, review manipulation also takes place in other forms, such as deleting or hiding unfavorable reviews⁹ and adding unfavorable reviews to others (Lappas et al. 2016, Luca and Zervas 2016, Nie et al. 2022). The costs and benefits of review manipulation vary according to the specific ways of review manipulation. For example, adding unfavorable reviews to other firms' sites and deleting unfavorable reviews are considered more unethical and riskier than self-injecting favorable reviews. Although it has been discovered that adding unfavorable reviews to potential competitors is more efficient in outperforming competitors (Lappas et al. 2016), deleting unfavorable online reviews is more effective in predicting product sales than self-injecting favorable online reviews because deleting negative reviews is more discrete and less likely to be noticed by consumers (Zhuang et al. 2018). Do firms establish mutual forbearance in other forms of review manipulation? Do the effects of rivalry on self-injecting favorable reviews differ from those of adding unfavorable reviews to others? Future studies should further explore the antecedents and consequences of different forms of review manipulation to illuminate similarities and differences. Second, although this study proposes the shift of collective regulatory focus and rivalry-induced deterrence as the theoretical underpinnings to explain the negative relationship between rivalry and self-injecting, these psychological underpinnings remain untested. Firms are encouraged to harness the benefits of rivalry and simultaneously avoid its downsides. Understanding the psychological underpinnings of rivalry in predicting unethical behaviors provides operable implications for reducing review manipulation (George et al. 2021, Kokkodis 2021, Yang et al. 2021, Naumzik et al. 2022). Therefore, exploring the conditions in which rivalry may be beneficial or harmful constitutes an essential next step. Third, we have examined the influence of rivalry on self-injecting favorable online reviews in the context of symmetrical rivalry, yet whether our findings can apply to asymmetric rivalry warrants further investigation. Rivalry is a subjective notion, such that one firm conceives the other as a rival; however, this perception may not be reciprocated (Kilduff et al. 2016, To et al. 2018).

Rivalry could be either asymmetric or symmetric. Future research can employ both observational and subjective data to identify asymmetric rivals and compare the effects of symmetric and asymmetric rivalry on firms' self-injecting behaviors. Fourth, it is worth noting that our measurement of rivals might not fully capture the dynamic nature of rivalry over time. Future research endeavors could focus on developing a more dynamic measurement approach to better capture the temporal variation in rivalry.

7.2. Conclusion

When consumers peruse a particular hotel on online reservation platforms, the platforms generally recommend a set of nearby alternatives. This recommendation system fuels the production of rivalry. Our theory-driven study suggests that the prevalence of online self-injecting behavior is notably lower in hotels experiencing stifling rivalry. This study suggests that while the presence of more rivals inhibits self-injecting, the self-injecting intensity among rivals stimulates self-injecting. These empirical findings portray rivalry as a multifaceted force, holding significant implications for policymakers, organizations, and e-platforms alike. It is expected that this study will spark additional research on rivalry vis-à-vis online review manipulation.

Endnotes

¹ Magic Johnson and Larry Bird were professional basketball players and key members of the Los Angeles Lakers and the Boston Celtics, respectively.

² Self-injecting behavior generally takes three forms: self-injecting positive reviews, deleting or hiding negative reviews, and posting negative reviews for competitors. This study focuses exclusively on self-injected positive reviews for two reasons. First, self-injected positive reviews are more ubiquitous than other forms of self-injecting. Most e-platforms do not allow sellers to hide or delete negative reviews about themselves. Posting negative reviews to competitors is less efficient in promoting a focal hotel's ranking. Hotels may not post negative reviews to all the competitors. Instead, they have to identify their major rivals (usually there is more than one) and post negative reviews for them. In case of some e-platforms, consumer can post a review only after they have purchased a product/service and made a payment. Unlike self-injected positive reviews—for which sellers hire people to buy their products with your own money (usually, the “hired” consumers just place an order and sellers do not ship the corresponding products) to post negative reviews to competitors—sellers have to hire people to buy the products of their competitors and make real payments. Therefore, posting negative reviews to competitors costs more than self-injected positive reviews, and the high cost makes fake negative reviews less ubiquitous. Second, self-inject positive reviews is more identifiable than the other two forms of self-injecting. Because sellers may post negative reviews for one or more competitors, we do not know who exactly the negative reviews were given to, which makes the act of posting negative reviews for competitors unmeasurable.

³ For example, hotels can report the unethical behaviors to authorities to seek punishment to their rivals. Reporting competitor's unethical behavior needs concrete evidence and the investigation

takes time. Even if the unethical firms are swiftly punished, firm's losses due to competitor's unethical behaviors cannot be subsidized.

⁴ The first round of telephone interview taken place in August 2023. Because hotel managers do not have any benefits to participate in our interview, the nonresponse rate is high. The missing responses include interviewees that hang up once we have made our intentions clear, interviewees that quitted the interview in the half; and the hotels that were not reached. Because our interview focuses on rivalry, a strategic topic of the hotels, responses from the hotel receptionist were excluded. After that, the first-round interview received 1,257 valid responses. To obtain more observations, we performed two more rounds of telephone interview that targeted the nonrespondents in the previous round. The second- and third-round interviews received 1,991 and 748 valid responses, respectively.

⁵ Based on our sample, 3,996 hotel managers participated in our phone interview (responses) and 2,985 hotels managers did not (nonresponses). We performed a Heckman two-stage model to examine whether there are systematic differences between the responses and nonresponses. Acknowledging the demanding workload shouldered by general managers, it is reasonable to anticipate a positive association between hotel size, review volume, and the variety of hotel facilities (e.g., business services, restaurant, swimming pool, fitness, etc.) and a manager's workload. Such workload may negatively predict their likelihood of participating in our interview (Chen et al. 2019, Forbes and Kosova 2022). In the first stage, we employed a Probit regression to estimate the probability that a hotel manager is likely to participate in our interview. The probability of participating in our interview is modeled as a function of the hotel size, hotel review volume, and variety of hotel facilities. In the second stage, we added the inverse mills ratio (IMR) estimated in the first stage to our main regression model. The IMR ($\beta = 0.012$, $p =$ not significant) was not significant in the second model, suggesting that self-selection bias does not appear to play a role.

⁶ Here we present a typical example to illustrate how subjective elements shape the identification process of rivals. Using objective measures, we initially identified 11 rivals for Shangri-La Hotel Pudong Shanghai: Peace Hotel, Park Hyatt, Grand Hyatt, The Ritz-Carlton, Wanda Reign, J Hotel, Max X Steigenberger, Riverside Hotel, Puyan Hotel, Blossom House, and IFCG Hotel. However, after conducting a thorough phone interview with the general manager, the number of identified rivals for Shangri-La Hotel Pudong Shanghai was reduced from 11 to 4. Specifically, the general manager selected three rivals (Park Hyatt, Grand Hyatt, The Ritz-Carlton) from the original list and nominated one additional rival, The Peninsula Shanghai. This discrepancy may be attributed to the relational and historical aspects of rivalry that were captured during the phone interview. These aforementioned statistics and examples effectively demonstrate the difference between rivalry and competition.

⁷ In the main analysis, we used $Rental_{i,t}$ as the instrumental of *Rivals*. We also used $Price_{i,t}$ as the instrumental of *Rivals* for robust check. The results of the 2SLS are identical.

⁸ The second condition for these IVs to be valid is that they should be uncorrelated with unobservable factors that may affect self-injecting intensity. In other words, $Rental_{i,t}$ and $Price_{i,t}$ should affect self-injecting intensity only through their effects on the *Rivals*. We test this exclusion restriction using the mediation analysis recommended by Zhao et al. (2010). We first regressed the IVs (commercial property price and commercial property rental) on our dependent variable, self-injecting intensity. The results show that both commercial property price and commercial property rental predict self-injecting intensity (commercial property prices: $\beta = 0.004$, $p < 0.050$; commercial property rental: $\beta = 0.007$, $p < 0.001$). Then we regressed both the IVs and the number of rivals on self-injecting intensity. The two IVs become insignificant (prices: $\beta = 0.003$, $p = 0.134$; rental: $\beta = -0.005$, $p = 0.114$). The mediation path is positive and significant

($\beta = 0.033$, the confidential interval [0.025,0.041] does not contain zero). Therefore, there are indirect-only effects of commercial property price and rental on self-injecting intensity, support the validity of using $Price_{i,t}$ and $Rental_{i,t}$ as the IVs.

⁹ Although it is unusual, evidence shows that firms do delete negative reviews in collaboration with the e-platforms. For example, in exchange for a monthly fee, Yelp allegedly offered to move a firm's negative reviews to the bottom of its Yelp page and make it difficult for users to access them (Morrison 2011).

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